

Matrosov's theorem for asymptotic stability, the time-invariant case

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Theorem 1 For the differential equation $\dot{x} = f(x)$, where f is continuous, the origin is globally asymptotically stable if both of the following conditions hold:

1. There exists a C^1 function $V_0 : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ that is positive definite and proper (i.e., radially unbounded) such that

$$Y_0(x) := \langle \nabla V_0(x), f(x) \rangle \leq 0 \quad \forall x \in \mathbb{R}^n; \quad (1)$$

2. There exist C^1 functions $V_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i \in \{1, \dots, m\}$, such that, with

$$Y_i(x) := \langle \nabla V_i(x), f(x) \rangle \quad \forall x \in \mathbb{R}^n, i \in \{1, \dots, m\}, \quad (2)$$

we have, for all $i \in \{1, \dots, m\}$,

$$x \in \mathbb{R}^n, Y_0(x) = \dots = Y_{i-1}(x) = 0 \implies Y_i(x) \leq 0 \quad (3)$$

and

$$x \in \mathbb{R}^n, Y_0(x) = \dots = Y_m(x) = 0 \implies x = 0. \quad (4)$$

Remark 1 If one assumes “global stability”, i.e., the existence of $\alpha \in \mathcal{K}_\infty$ such that $|x(t)| \leq \alpha(|x(0)|)$ for all solutions $x(\cdot)$ and all $t \geq 0$, then there is no need to assume that V_0 is nonnegative (let alone, positive definite) nor proper in item 1 above. ■

Theorem 2 For the differential equation $\dot{x} = f(x)$, where f is continuous, the origin is locally asymptotically stable if the conditions of Theorem 1 hold with every instance of $\mathbb{R}^n$ replaced by a neighborhood $U \subset \mathbb{R}^n$ of the origin and without needing to verify that V_0 is proper.

Example 1 Consider the system with state $x := (z^T, v)^T$ where $z \in \mathbb{R}^n$ and $v \in \mathbb{R}$ given by

$$\dot{z} = Az + Bv \quad (5a)$$

$$\dot{v} = -h(v) - Cz \quad (5b)$$

where $h : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, $vh(v) > 0$ for all $v \neq 0$, (C, A) is detectable, and there exists $P = P^T > 0$ such that $A^T P + PA \leq 0$ and $PB = C^T$. To establish GAS via Theorem 1,

0. Define $V_0(x) = z^T Pz + v^2$. It is clear that V_0 is positive definite and radially unbounded. It is not difficult to establish, using $A^T P + PA \leq 0$ and $PB = C^T$ that

$$Y_0(x) := \langle \nabla V_0(x), f(x) \rangle \leq -vh(v) \leq 0.$$

Thus, the first condition of Theorem 1 holds. It is also useful to observe that $Y_0(x) = 0$ implies $v = 0$. This means that, as we consider subsequent functions V_1 , etc., we do not need worry about terms in their derivatives that vanish when $v = 0$.

1. Next, define $V_1(x) := z^T C^T v$. Then, we have

$$Y_1(x) := \langle \nabla V_1(x), f(x) \rangle = -z^T C^T (-h(v) - Cz) + (Az + Bv)^T C^T v.$$

Notice that $Y_0(x) = 0$ implies $v = 0$ which implies $Y_1(x) = -z^T C^T Cz \leq 0$. Notice also that $Y_0(x) = Y_1(x) = 0$ implies $v = 0$ and $Cz = 0$. This means that, as we consider subsequent functions V_2 , etc., we do not need to worry about terms in their derivatives that vanish when $v = 0$ or $Cz = 0$.

2. Next, define $V_2(x) := z^T Xz$ where $X = X^T > 0$ and

$$(A - LC)^T X + X(A - LC) = -I$$

where L is such that $A - LC$ is Hurwitz. These matrices exist since (C, A) is assumed to be detectable. We get

$$Y_2(x) := \langle \nabla V_2(x), f(x) \rangle = -z^T z + 2z^T X(LCz + Bv).$$

Notice that $Y_0(x) = Y_1(x) = 0$ implies $v = 0$ and $Cz = 0$, which implies that $Y_2(z) = -z^T z \leq 0$. Also, $Y_0(x) = Y_1(x) = 0 = Y_2(x) = 0$ implies that $v = 0$ and $z = 0$.

Thus, the conditions of Theorem 1 hold with $m = 2$ and so the origin of the system (5) is GAS.